

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 2

Total Marks: 15

Annexure A

CONCEPT MAPPING

Code of Conduct:

I **M. Nishanth (192511154)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Nishanth

Annexure A

CONCEPT MAPPING

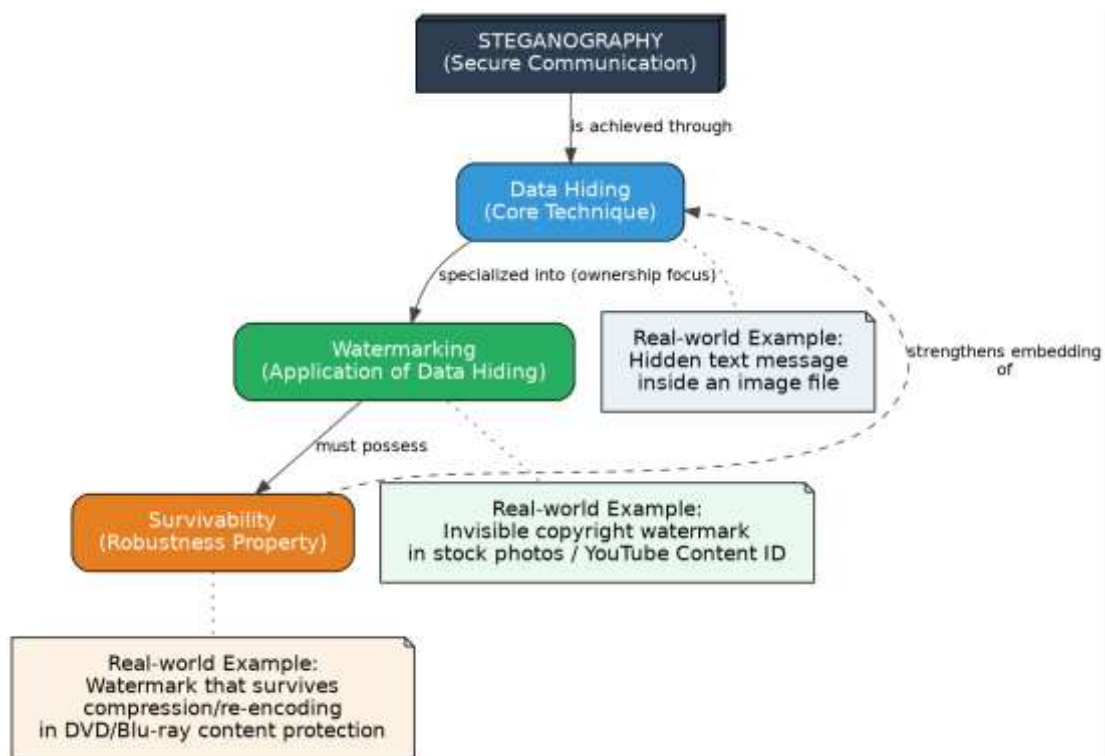
1. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each

2. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

Question 1: Concept Map Steganography (Data Hiding, Watermarking, Survivability)

Task: Create a concept map connecting Data Hiding, Watermarking and Survivability; show the relationships between these concepts, analyze how each contributes to secure communication, and include one real-world application for each.

Concept Map



a. Relationships Between the Concepts

The three concepts form a hierarchical and mutually reinforcing structure within the broader discipline of steganography and information hiding:

1. Data Hiding is the foundational, general-purpose concept: it refers to the process of embedding secret information imperceptibly within a cover medium (image, audio, video, or text) so that the very existence of the hidden data is concealed from an observer.
2. Watermarking is a specialized application of data hiding. Rather than hiding an arbitrary secret message for covert communication, watermarking embeds identifying information (such as an owner's signature, a copyright notice, or a serial number) into a cover object, primarily to prove ownership, authenticate origin, or track unauthorized distribution.

3. Survivability (also called robustness) is a required property layered on top of both data hiding and watermarking. It measures whether the embedded information remains recoverable after the cover object undergoes common transformations — such as compression, cropping, resizing, noise addition, or format conversion. Watermarking in particular depends heavily on survivability, because a watermark that can be trivially removed by re-saving or resizing an image provides no real copyright protection.
4. In short: Data Hiding → generalizes to → Watermarking (as an ownership-focused sub-technique) → which must possess → Survivability (to remain effective under real-world processing and attack). Survivability, in turn, feeds back into how the original data-hiding embedding algorithm must be designed (e.g., embedding in perceptually robust regions of an image rather than fragile ones).

b. Contribution of Each Concept to Secure Communication

1. Data Hiding contributes secrecy: it enables covert communication channels where the presence of a hidden message is not obvious to an eavesdropper, which is useful when even the existence of communication must be concealed (unlike encryption, which conceals content but not the fact that a secret message exists).
2. Watermarking contributes authenticity and accountability: it allows a receiver, auditor, or automated system to verify the origin or ownership of digital content, deterring piracy, unauthorized redistribution, and tampering, and supporting non-repudiation in digital rights management.
3. Survivability contributes reliability and trustworthiness of the security mechanism itself: without robustness against routine processing (which is not even a deliberate attack), any embedded secret or ownership mark would be lost, defeating the purpose of the hiding or watermarking scheme. Survivability is therefore what allows the security guarantee to hold in practical, real-world conditions rather than only in a laboratory setting.

c. Real-World Applications

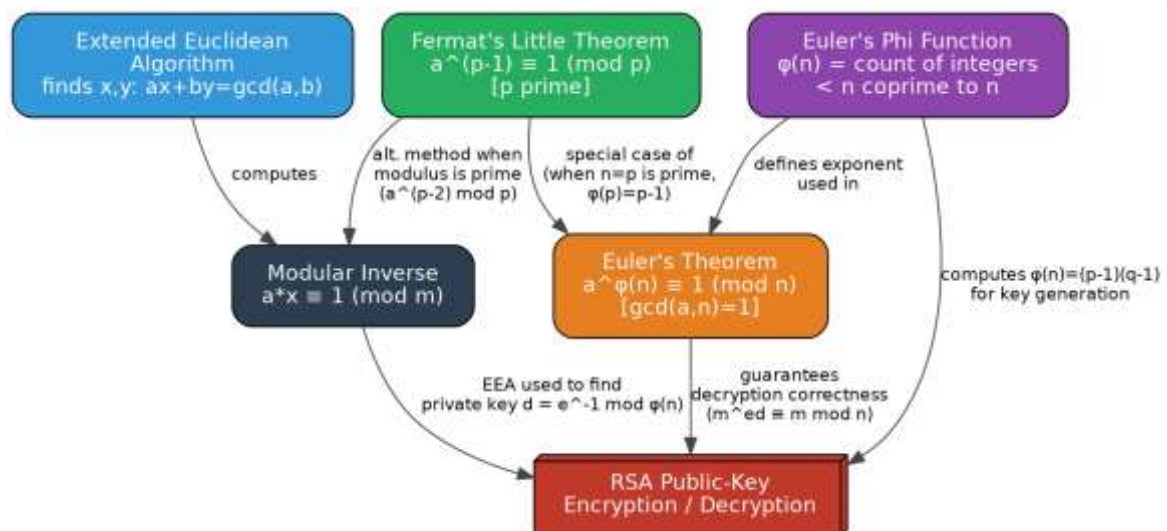
1. Data Hiding: Covert steganographic communication — for example, embedding a secret text message inside the least significant bits of an image file, used historically for confidential communication that must not draw attention (e.g., in espionage or secure journalism to protect sources).

2. Watermarking: Digital Rights Management (DRM) systems such as YouTube's Content ID or invisible watermarks embedded in stock photography (e.g., Shutterstock/Getty Images previews), which automatically identify copyrighted material and its rights holder even after the content has been copied or re-uploaded.
3. Survivability: Broadcast and disc-based content protection (e.g., DVD/Blu-ray or cinema anti-piracy watermarks) engineered to survive re-encoding, compression, camcorder re-recording, and format conversion, allowing forensic tracing of leaked or pirated copies back to the original distribution source.

Question 2: Concept Map – Number Theory in Cryptography

Task: Draw a concept map linking Modular Inverse, Extended Euclidean Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem; show their mathematical connections, map their role in encryption/decryption, and analyze which concept is most critical for public-key cryptography.

Concept Map



a. Mathematical Connections

1. Extended Euclidean Algorithm (EEA): Given integers a and m , the EEA finds integers x and y such that $a \cdot x + m \cdot y = \gcd(a, m)$. When $\gcd(a, m) = 1$ (a and m are coprime), this equation directly yields x as the modular inverse of a modulo m , since $a \cdot x \equiv 1 \pmod{m}$. The EEA is therefore the general-purpose, constructive method for computing modular inverses for any modulus, prime or composite.

2. **Modular Inverse:** The modular inverse of a modulo m is the integer x satisfying $a \cdot x \equiv 1 \pmod{m}$. It exists if and only if $\gcd(a, m) = 1$, and can be computed either via the Extended Euclidean Algorithm (general case) or, when the modulus is a prime p , via Fermat's Little Theorem as $x = a^{(p-2)} \pmod{p}$.
3. **Fermat's Little Theorem (FLT):** States that if p is a prime number and a is not divisible by p , then $a^{(p-1)} \equiv 1 \pmod{p}$. This is a special case of Euler's Theorem that applies only when the modulus is prime, because for a prime p , Euler's Phi Function gives $\phi(p) = p - 1$.
4. **Euler's Phi Function (ϕ):** Counts the number of positive integers less than n that are coprime to n . For a prime p , $\phi(p) = p - 1$; for the product of two distinct primes $n = p \cdot q$ (as used in RSA), $\phi(n) = (p - 1)(q - 1)$. This function defines the exponent used in Euler's Theorem and is the basis for computing the totient needed in RSA key generation.
5. **Euler's Theorem:** A generalization of Fermat's Little Theorem to any modulus n (not just primes): if $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$. This theorem is the fundamental mathematical guarantee underpinning RSA's correctness — it is what proves that decrypting an RSA-encrypted ciphertext correctly recovers the original plaintext.

b. Role in Encryption / Decryption (RSA)

These five concepts combine directly to enable RSA public-key encryption:

1. **Key Generation:** Two large primes p and q are chosen and multiplied to form the modulus $n = p \cdot q$. Euler's Phi Function computes $\phi(n) = (p-1)(q-1)$, which is needed for the next step.
2. **An encryption exponent e is chosen such that $\gcd(e, \phi(n)) = 1$.** The Extended Euclidean Algorithm is then used to compute the Modular Inverse of e modulo $\phi(n)$, giving the private decryption exponent d , i.e., $d = e^{-1} \pmod{\phi(n)}$.
3. **Encryption/Decryption:** A message m is encrypted as $c = m^e \pmod{n}$, and decrypted as $m = c^d \pmod{n}$. The correctness of this round trip — that decrypting the ciphertext really does return the original message — is mathematically guaranteed by Euler's Theorem (with Fermat's Little Theorem as the underlying special-case intuition when reasoning about the prime factors p and q individually via the Chinese Remainder Theorem).

c. Most Critical Concept for Public-Key Cryptography (with Justification)

Among the five, Euler's Theorem (together with the Euler's Phi Function that defines its exponent) is arguably the single most critical concept for public-key cryptography:

1. It is the direct mathematical proof of correctness for RSA — without $a^{\varphi(n)} \equiv 1 \pmod{n}$ holding true, there would be no guarantee that decryption ever recovers the original plaintext, meaning RSA simply would not work as an encryption scheme.
2. It generalizes Fermat's Little Theorem to composite moduli, which is essential because RSA's modulus $n = p \cdot q$ is always composite by design; a special case that only worked for prime moduli (FLT alone) would be insufficient for RSA.
3. Euler's Phi Function, which Euler's Theorem depends on, is also what makes the security of RSA possible in the first place: computing $\varphi(n)$ is computationally easy only if the prime factors p and q are known, and hard (equivalent to factoring n) otherwise — this asymmetry between easy key generation (by the legitimate holder of p and q) and hard key recovery (by an attacker who only sees n) is the entire basis of RSA's public-key security model.
4. While the Extended Euclidean Algorithm and Modular Inverse are indispensable computational tools (they are how the private key d is actually calculated in practice), they are supporting mechanisms; they compute a value whose correctness and security significance is established by Euler's Theorem and the Phi Function. In that sense, Euler's Theorem is the theoretical foundation, while EEA/Modular Inverse and Fermat's Little Theorem are the practical and special-case tools built upon that foundation.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation